

MANDAR BIREWAR

Nagpur, Maharashtra, India

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Education

Shri Ramdeobaba College of Engineering and Management

Nagpur, India

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

– CGPA: 8.23

Taywade College

Nagpur, India

Class XII – State Board

2023

– Percentage: 81.5%

Bhavan's BP Vidya Mandir, Ashti

Nagpur, India

Class X – CBSE Board

2021

– Percentage: 83%

Projects

AI Interview Copilot | FastAPI, LangChain, Mistral AI, ChromaDB

Jun 2026

- Developed a Retrieval-Augmented Generation (RAG) platform that generates personalized interview questions from uploaded resumes using Mistral AI, LangChain, and ChromaDB.
- Built FastAPI-based APIs for resume ingestion, semantic retrieval, vector storage, and context-aware interview assistance.

DocuMind AI | FastAPI, Hugging Face, ChromaDB, RAG

May 2026

- Built a full-stack AI-powered PDF assistant enabling semantic search, document summarization, and question answering using Retrieval-Augmented Generation (RAG).
- Implemented PDF parsing, embedding generation, vector search using ChromaDB and Hugging Face models, and a responsive chat-based interface.

HVAC Leak Prediction System | XGBoost, LSTM, PyTorch, Computer Vision

May 2026

- Developed an AI-powered predictive maintenance system for HVAC leak-risk prediction using sensor analytics, anomaly detection, and risk classification.
- Engineered features from HVAC datasets and implemented XGBoost and LSTM models for fault detection and maintenance recommendations.

AI Deepfake & Image Forgery Detection | Python, PyTorch, OpenCV

Mar 2026

- Developed a CNN-based system for detecting manipulated and cut-paste images using deep learning techniques.
- Implemented Grad-CAM visualization and evaluated model performance using accuracy, precision, recall, and ROC-AUC metrics.

Technical Skills

Programming Languages: Python, C++, SQL

Generative AI: LangChain, RAG, ChromaDB, Hugging Face, Mistral AI

Machine Learning & Data Science: Scikit-Learn, Pandas, NumPy, Matplotlib, PyTorch, OpenCV

Backend & Tools: FastAPI, Git, GitHub, VS Code, Jupyter Notebook

Computer Science Fundamentals: DSA, OOPS, DBMS, Operating Systems, Computer Networks

Achievements

- Winner, Orange Promptathon 2026 (Team SupaStrikas), organized by Ramdeobaba University in collaboration with Carrier HVAC, InfoCepts, and NASSCOM, for developing an AI-Powered Predictive Leak Intelligence System using Computer Vision, XGBoost, LSTM, and Digital Twin simulation.